

WAYNE, NE, USA

WMO: 722241

Lat: 42.242N

Lon: 96.983W

Elev: 1424

StdP: 13.96

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.8	(c) -1.4	(d) -13.3	(e) 2.8	(f) -6.0	(g) -8.4	(h) 3.7	(i) -0.4	(j) 31.9	(k) 26.4	(l) 28.6	(m) 23.5	(n) 5.3	(o) 320	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	93.0	72.9	89.9	72.5	87.1	71.6	78.5	85.7	76.8	84.5	75.0	83.0	14.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	147.2	83.1	74.4	135.7	81.4	72.5	126.9	79.9	43.2	85.6	41.2	84.8	39.4	82.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.8	21.7	-18.0	97.1	5.5	3.2	-21.9	99.4	-25.2	101.3	-28.2	103.0	-32.2	105.3
			-18.3	81.9	5.4	2.2	-22.3	83.5	-25.4	84.8	-28.5	86.0	-32.4	87.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	21.3	23.3	37.8	47.9	59.8	70.5	73.8	70.7	63.9	49.7	36.9	24.1
	DBStd	21.07	12.50	12.37	12.07	9.97	9.12	6.53	6.07	5.36	8.26	9.60	10.38	11.26
	HDD50	3553	889	748	409	153	18	0	0	0	5	123	405	803
	HDD65	6853	1354	1168	844	518	214	22	7	13	119	484	842	1268
	CDD50	2985	0	1	32	90	322	616	737	641	420	113	13	0
	CDD65	809	0	0	1	5	53	188	279	189	84	10	0	0
	CDH74	8592	0	0	36	131	708	2017	2982	1674	903	137	4	0
	CDH80	3110	0	0	5	42	255	738	1207	534	283	46	0	0
Wind	WSAvg	9.1	9.6	10.0	10.7	11.5	10.6	8.8	6.7	6.3	7.9	8.9	9.4	9.2
Precipitation	PrecAvg	28.2	0.7	0.8	2.1	2.9	4.1	4.4	3.3	2.8	2.9	2.0	1.5	0.8
	PrecMax	40.7	2.0	3.4	6.6	11.2	9.1	12.4	10.3	7.9	9.6	6.3	4.5	3.0
	PrecMin	16.3	0.0	0.0	0.2	0.4	1.3	1.0	0.4	0.5	0.3	0.0	0.0	0.0
	PrecStd	5.8	0.4	0.7	1.6	2.0	1.7	2.3	2.0	1.7	1.8	1.5	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	62.0	79.8	86.2	92.1	95.3	98.8	94.7	91.8	86.5	72.3	55.2
		MCWB	44.1	48.6	59.2	60.8	66.6	72.5	73.5	73.0	70.3	66.7	55.4	45.4
	2%	DB	46.6	51.7	71.2	77.1	87.2	90.8	94.1	89.0	87.0	76.9	65.4	49.7
		MCWB	39.3	43.3	54.0	56.7	65.5	71.9	74.5	74.4	70.4	59.9	52.1	42.4
	5%	DB	41.1	45.1	63.7	71.2	81.7	87.8	90.1	86.0	82.8	72.1	59.1	43.3
		MCWB	36.6	39.2	53.3	55.3	63.2	70.2	75.7	73.5	68.2	58.2	48.1	38.0
	10%	DB	37.1	39.7	56.9	66.2	77.1	84.2	86.6	82.3	79.4	66.9	53.6	38.6
		MCWB	34.2	36.2	48.4	52.7	61.8	69.1	74.0	71.7	66.7	55.8	45.4	34.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	51.2	62.5	63.9	72.3	79.2	82.3	79.5	76.7	69.0	58.2	48.2
		MCDB	55.3	63.5	72.4	77.2	82.6	86.4	88.0	86.4	83.4	79.4	67.7	52.7
	2%	WB	40.1	43.5	58.1	60.5	69.3	76.3	79.6	77.5	74.0	64.3	53.4	42.0
		MCDB	45.6	50.7	66.1	71.6	80.0	85.1	86.3	84.4	81.5	73.1	63.4	47.4
	5%	WB	37.0	39.7	53.5	57.6	66.3	74.0	77.5	75.7	71.8	60.6	49.6	38.5
		MCDB	40.8	44.7	63.2	68.1	76.6	82.7	85.3	82.7	79.4	68.2	58.1	43.4
	10%	WB	34.4	36.1	48.8	54.6	63.7	72.1	75.5	73.8	69.4	56.6	45.7	35.4
		MCDB	37.2	39.8	58.2	64.0	74.1	80.7	84.0	81.0	76.7	65.8	53.3	38.4
Mean Daily Temperature Range	5% DB	MDBR	18.7	19.1	21.4	23.8	23.5	22.3	22.2	21.1	24.9	24.4	23.1	17.8
		MCDBR	23.3	25.6	33.9	34.1	30.4	25.8	24.9	24.9	27.8	32.4	32.0	24.3
		MCWBR	16.6	18.0	19.2	17.2	13.6	10.9	10.7	12.6	13.0	17.0	19.1	17.8
	5% WB	MCDBR	22.2	24.5	30.0	27.3	24.3	20.8	21.1	20.8	21.5	25.4	29.6	22.2
		MCWBR	16.3	17.8	18.0	15.5	12.8	11.5	12.0	12.6	12.6	15.3	19.0	16.7
Clear-Sky Solar Irradiance	taub		0.274	0.290	0.317	0.389	0.411	0.418	0.424	0.421	0.371	0.323	0.287	0.278
	taud		2.448	2.436	2.417	2.223	2.237	2.278	2.268	2.270	2.373	2.493	2.507	2.454
	Ebn at Noon		280	292	295	279	274	272	269	266	273	276	273	268
	Edh at Noon		23	28	33	43	43	42	42	41	34	27	22	22
All-Sky Solar Radiation	RadAvg		626	944	1224	1564	1833	2056	2101	1774	1436	1000	698	514
	RadStd		54	72	98	110	135	148	108	129	98	118	49	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page